

Output 1

Enter string: test

PREFIXES:

t
te
tes
test

SUFFIXES:

test
est
st
t

SUBSTRINGS:

t
te
tes
test
e
es
est
s
st
t

Name: Samir Paudel

Roll No: 114/079

Lab No: 1

○ [sam@arch samir_paudel]\$

Output 2

```
• [sam@arch samir_paudel]$ cd "/home/sam/Github/semester-wc
ler-design/samir_paudel/" && g++ lab2_keyword.cpp -o lab2
am/Github/semester-works/semester-6/compiler-design/samir
d
Enter string: int
int is a KEYWORD

Name: Samir Paudel
Roll No: 114/079
Lab No: 2
• [sam@arch samir_paudel]$ cd "/home/sam/Github/semester-wc
ler-design/samir_paudel/" && g++ lab2_keyword.cpp -o lab2
am/Github/semester-works/semester-6/compiler-design/samir
d
Enter string: samir
samir is a valid IDENTIFIER

Name: Samir Paudel
Roll No: 114/079
Lab No: 2
○ [sam@arch samir_paudel]$ █
```

Output 3

DFA Menu:

1. Starts with 01
2. Ends with 010
3. Contains 011

Enter choice: 1

Enter string (0s and 1s): 010111

ACCEPTED

DFA Menu:

1. Starts with 01
2. Ends with 010
3. Contains 011

Enter choice: 2

Enter string (0s and 1s): 110010

ACCEPTED

DFA Menu:

1. Starts with 01
2. Ends with 010
3. Contains 011

Enter choice: 3

Enter string (0s and 1s): 010101

REJECTED

Name: Samir Paudel

Roll No: 114/079

Lab No: 3

[sam@arch samir_paudel]\$

* History restored

Output 4

```
Enter string (a's and b's): aabaa  
REJECTED (Even number of a's)
```

```
Name: Samir Paudel
```

```
Roll No: 114/079
```

```
Lab No: 4
```

```
○ [sam@arch samir_paudel]$ █
```

Output 5

Enter a line: Single line comment
Not a comment

Name: Samir Paudel

Roll No: 114/079

Lab No: 5

○ [sam@arch samir_paudel]\$ █

Output 6

```
Enter number of productions: 2
Enter productions (e.g., A=aB):
S=aA
```

```
A=b
Enter symbol to find FIRST: S
FIRST(S) = { a }
```

```
Name: Samir Paudel
```

```
Roll No: 114/079
```

```
Lab No: 6
```

```
○ [sam@arch samir_paudel]$
```

Output 7

```
Enter number of productions: 2
Enter productions (e.g., S=AB):
S=AB
A=a
Enter symbol to find FOLLOW: A
FOLLOW(A) = { B }
```

Name: Samir Paudel

Roll No: 114/079

Lab No: 7

○ [sam@arch samir_paudel]\$ █

Output 8

- [sam@arch samir_paudel]\$ cd "/home/sam/Github/semest
sign/samir_paudel/" && g++ lab8_identifier.cpp -o la
hub/semester-works/semester-6/compiler-design/samir_
Enter identifier: _variable
_variable is VALID identifier

Name: Samir Paudel

Roll No: 114/079

Lab No: 8

- [sam@arch samir_paudel]\$ cd "/home/sam/Github/semest
sign/samir_paudel/" && g++ lab8_identifier.cpp -o la
hub/semester-works/semester-6/compiler-design/samir_
Enter identifier: 123abc
123abc is INVALID identifier

Name: Samir Paudel

Roll No: 114/079

Lab No: 8

- [sam@arch samir_paudel]\$ █

1

Output 9

- [sam@arch samir_paudel]\$ cd "/home/sam/Github/semester-works/semester-6/compiler-design/samir_paudel/" && g++ lab9_operators.cpp -o lab9_operators && "/home/b/semester-works/semester-6/compiler-design/samir_paudel/"lab9_operators
Enter operator: +
+ is a valid OPERATOR

Name: Samir Paudel
Roll No: 114/079
Lab No: 9

- [sam@arch samir_paudel]\$ cd "/home/sam/Github/semester-works/semester-6/compiler-design/samir_paudel/" && g++ lab9_operators.cpp -o lab9_operators && "/home/b/semester-works/semester-6/compiler-design/samir_paudel/"lab9_operators
Enter operator: -
- is a valid OPERATOR

Name: Samir Paudel
Roll No: 114/079
Lab No: 9

- [sam@arch samir_paudel]\$ cd "/home/sam/Github/semester-works/semester-6/compiler-design/samir_paudel/" && g++ lab9_operators.cpp -o lab9_operators && "/home/b/semester-works/semester-6/compiler-design/samir_paudel/"lab9_operators
Enter operator: 3
3 is NOT a valid OPERATOR

Name: Samir Paudel
Roll No: 114/079
Lab No: 9

- [sam@arch samir_paudel]\$

Output 10

```
Enter number of symbols: 3
Enter symbols (name address type):
x 1000 int
y 2000 float
z 3000 char
```

Symbol Table:

Name	Address	Type
x	1000	int
y	2000	float
z	3000	char

```
Enter symbol to search: x
Found: x at address 1000, type int
```

```
Name: Samir Paudel
Roll No: 114/079
Lab No: 10
```

```
○ [sam@arch samir_paudel]$
```

Output 11

Grammar production ($E=E+E$, $E=id$):
Enter input string: id+id

Stack	Input	Action
\$	id+id\$	Shift i
\$i	d+id\$	Shift d
\$E	+id\$	Reduce id->E
\$E	+id\$	Shift +
\$E+	id\$	Shift i
\$E+i	d\$	Shift d
\$E+E	\$	Reduce id->E
\$E+E	\$	Shift \$

Name: Samir Paudel
Roll No: 114/079
Lab No: 11

○ [sam@arch samir_paudel]\$

Output 12

```
1.insert 2.display 3.delete 4.search 5.modify 6.exit
choice: 1
enter label: loop
enter address: 2000
inserted
```

```
1.insert 2.display 3.delete 4.search 5.modify 6.exit
choice: 2
```

```
label    address
loop     2000
```

```
1.insert 2.display 3.delete 4.search 5.modify 6.exit
choice: 4
enter label to search: loop
found at address: 2000
```

```
1.insert 2.display 3.delete 4.search 5.modify 6.exit
choice: 3
enter label to delete: loop
deleted
```

```
1.insert 2.display 3.delete 4.search 5.modify 6.exit
choice: 2
symbol table empty
```

```
1.insert 2.display 3.delete 4.search 5.modify 6.exit
choice: 6
```

```
name: samir paudel
roll no: 114/079
lab no: 12
```

```
○ [sam@arch samir_paudel]$ █
```

Output 13

```
• [sam@arch samir_paudel]$ cd "/home/sam/Github/semester-works/semester-6/comp
  ign/samir_paudel/" && g++ lab13_tac.cpp -o lab13_tac && "/home/sam/Github/!
  orks/semester-6/compiler-design/samir_paudel/"lab13_tac
  Enter infix expression: a+b*c
  Postfix: abc*+
```

Three Address Code:

```
*, b, c, t1
+, a, t1, t2
```

Name: Samir Paudel

Roll No: 114/079

Lab No: 13

```
• [sam@arch samir_paudel]$ cd "/home/sam/Github/semester-works/semester-6/comp
  ign/samir_paudel/" && g++ lab13_tac.cpp -o lab13_tac && "/home/sam/Github/!
  orks/semester-6/compiler-design/samir_paudel/"lab13_tac
  Enter infix expression: a*b+c*d
  Postfix: ab*cd*+
```

Three Address Code:

```
*, a, b, t1
*, c, d, t2
+, t1, t2, t3
```

Name: Samir Paudel

Roll No: 114/079

Lab No: 13

```
○ [sam@arch samir_paudel]$ █
```

Output 14

```
• [sam@arch samir_paudel]$ cd "/home/sam/Github/semest
ign/samir_paudel/" && g++ lab14_optimizer.cpp -o l
b/semester-works/semester-6/compiler-design/samir_
Enter number of TAC statements: 4
Enter TAC (op arg1 arg2 result):
= x _ t1
= y _ t2
+ t1 t2 t3
= z _ t4

Optimized TAC:
= x _ t1
= y _ t2
+ t1 t2 t3

Name: Samir Paudel
Roll No: 114/079
Lab No: 14
○ [sam@arch samir_paudel]$ █
```